

$$Z[1(t)]$$

$$= \sum_{n=0}^{\infty} e(nT) \cdot z^{-n}$$

$$= 1 \cdot z^0 + 1 \cdot z^{-1} + 1 \cdot z^{-2} + \dots$$

$$= E(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$$

$$P = \frac{238,6}{s^2 + 10,025s} = \frac{238,6}{s(s+10,025)} = \frac{A}{s} + \frac{B}{s+10,025}$$

$$\Rightarrow As + 10,025A + Bs$$

$$= (A+B)s + 10,025A = 238,6$$

$$\Rightarrow \begin{cases} A+B = 0 \\ 10,025A = 238,6 \end{cases}$$

$$\Rightarrow \begin{cases} A = 23,8 \\ B = -23,8 \end{cases}$$

$$\begin{aligned} \therefore P &= \frac{23,8}{s} + \frac{-23,8}{s+10,025} = 23,8 \left(\frac{z}{z-1} - \frac{z}{z - e^{-10,025T}} \right) \\ &= 23,8 \left(\frac{z}{z-1} - \frac{z}{z-0,9046} \right) \\ &= 23,8 z \left(\frac{z-0,9046 - z + 1}{(z-1)(z-0,9046)} \right) \\ &= 23,8 z \left(\frac{0,0954}{z^2 - 1,9046z + 0,9046} \right) \end{aligned}$$

$$\text{Pole: } z_1 = e^{10T} = 0,3679$$

$$\text{Zero: } z_1 = e^{-10T} = 0,9048$$

$$D_d = \frac{z - 0,9048}{z - 0,3679}$$

$$2.2 \times \frac{100}{100} = K_d \cdot \frac{1-0.9048}{1-0.3679}$$

$$\Rightarrow K_d = 14.607$$

$$E(t) = \sum_{k=0}^{\infty} e_k z^k = \sum_{k=0}^{\infty} K T \cdot z^{-k}$$

$$= 0 + T \cdot z^{-1} + 2T \cdot z^{-2}$$

$$= T z^{-1} (1 + 2z^{-1} + 3z^{-2} + \dots)$$

$$= T z^{-1}$$

Binomial series

$$\begin{cases} x = -z^{-1} \\ \alpha = -2 \end{cases}$$

$$y(k+1) - y(k) = u(k+1) + u(k)$$

$$Y(z) \cdot z - Y(z) = U(z) \cdot z + U(z)$$

$$\frac{Y}{U} = \frac{z+1}{z-1}$$

$$Y(z) = \frac{z+1}{z-1} \cdot \frac{T \cdot z}{(z-1)^2}$$

$$= \frac{(z+1)zT}{(z-1)^3}$$

$$= \frac{Tz^2}{(z-1)^3} + \frac{zT}{(z-1)^3}$$

$$2.2 \cdot \frac{1054/100}{8+100} = K_d \cdot = 0.551$$

$$z_1 = e^{-10T} = 0.0703$$

$$z_2 = e^{-100T} = 0.0183$$

$$D_d = \frac{z - 0.6703}{z - 0.0183}$$